Medication Related Osteonecrosis Of The Jaws

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HEALTH SCIENCES CENTRE
when it matters MOST

Outline

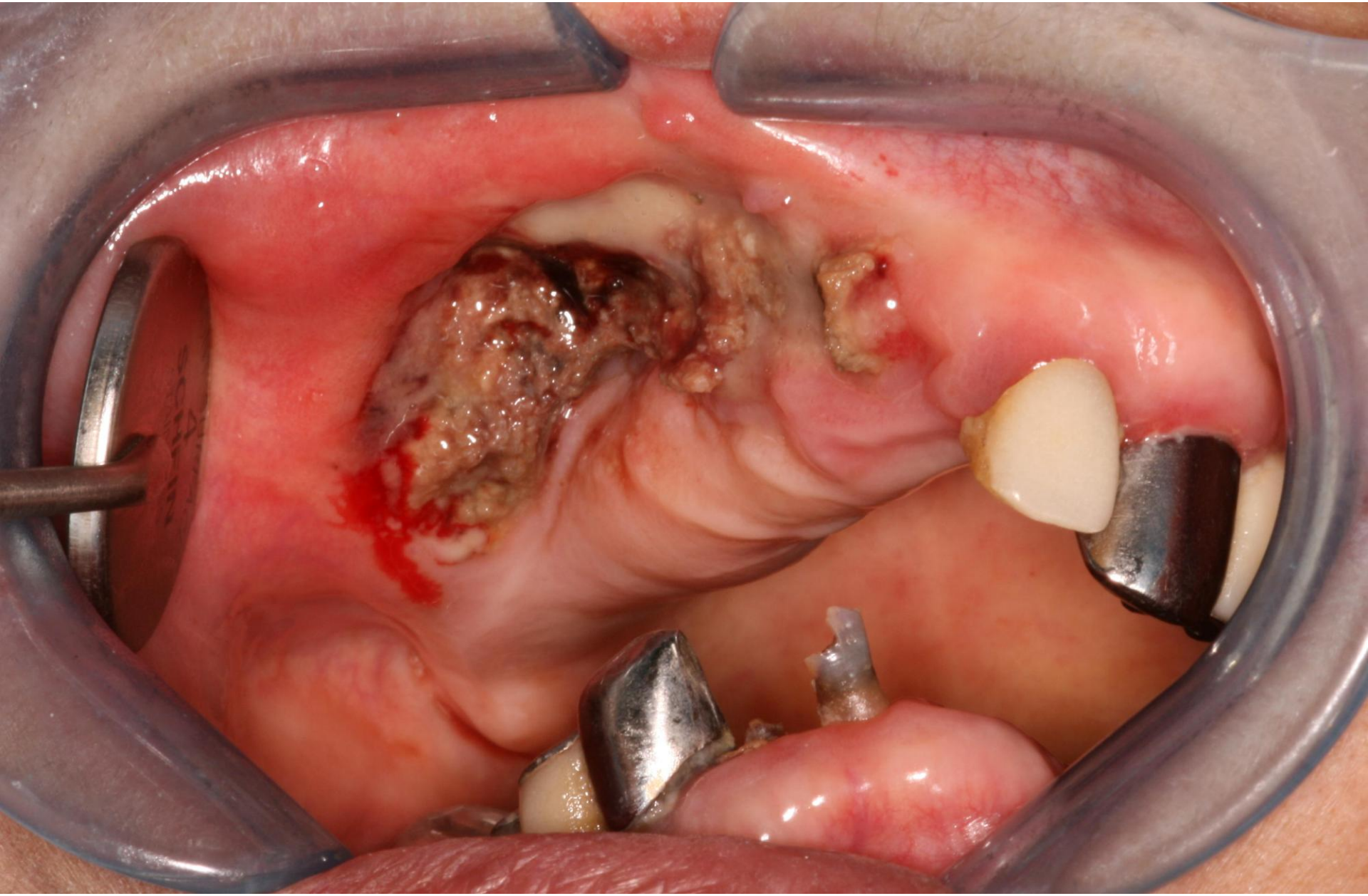
- Definition.
- History.
- Bisphosphonates.
- RANKL Inhibitors.
- Staging.
- Differential diagnosis.
- Incidence.

Outline

- Risk factors.
- Initiating factors.
- Vulnerable sites.
- Co-morbidities.
- The cancer patient.
- The osteoporotic patient.

Definition

- Bisphosphonate Related Osteonecrosis of the Jaws “BRONJ”:
  - Exposed bone in the maxillofacial area for more than 8 weeks, in the absence of head and neck radiation, in a patient on bisphosphonate therapy in the present or past.



Definition

- Medication Related Osteonecrosis of the Jaws “MRONJ”:
  - “Induced” vs “Related”.
  - “Medication” vs “Drug”.
  - Non-bisphosphonate drugs.

J Oral Maxillofac Surg September 2003: 61 (9)

Wang et al (UCSF)

Osteonecrosis of the jaws associated with cancer chemotherapy

Marx (U. Miami)

Pamidronate (Aredia) and zoledronate (Zometa) induced avascular necrosis of the jaws: a growing epidemic

Wang et al (UCSF) **Osteonecrosis of the jaws associated with cancer chemotherapy**

- 3 case reports
- “first reported cases of osteonecrosis secondary to cancer chemotherapy with drugs in the taxoid family”
- “all 3 patients were also treated with Pamidronate, but this medication is used to inhibit bone resorption and has not previously been associated with bone necrosis”

Marx (U. Miami) **Pamidronate (Aredia) and zoledronate (Zometa) induced avascular necrosis of the jaws: a growing epidemic**

- 36 cases reported in letter to editor
- "painful bone exposure in the mandible, maxilla, or both, that

  were unresponsive to surgical or

  medical treatments" related to

  either Pamidronate or zoledronate

J Oral Maxillofac Surg May, 2004: 62(5)

Ruggiero et al (Long Island Jewish Medical Center) **Osteonecrosis of the jaws associated with the use of bisphosphonates: a review of 63 cases**

Drugs reported to cause MRONJ

- Bisphosphonates.
- RANKL inhibitors.
- Antiangiogenic drugs.

Bisphosphonates

Indications for Bisphosphonate Therapy

Metabolic bone disease

- Osteoporosis, Paget's disease, osteogenesis imperfecta

Cancer

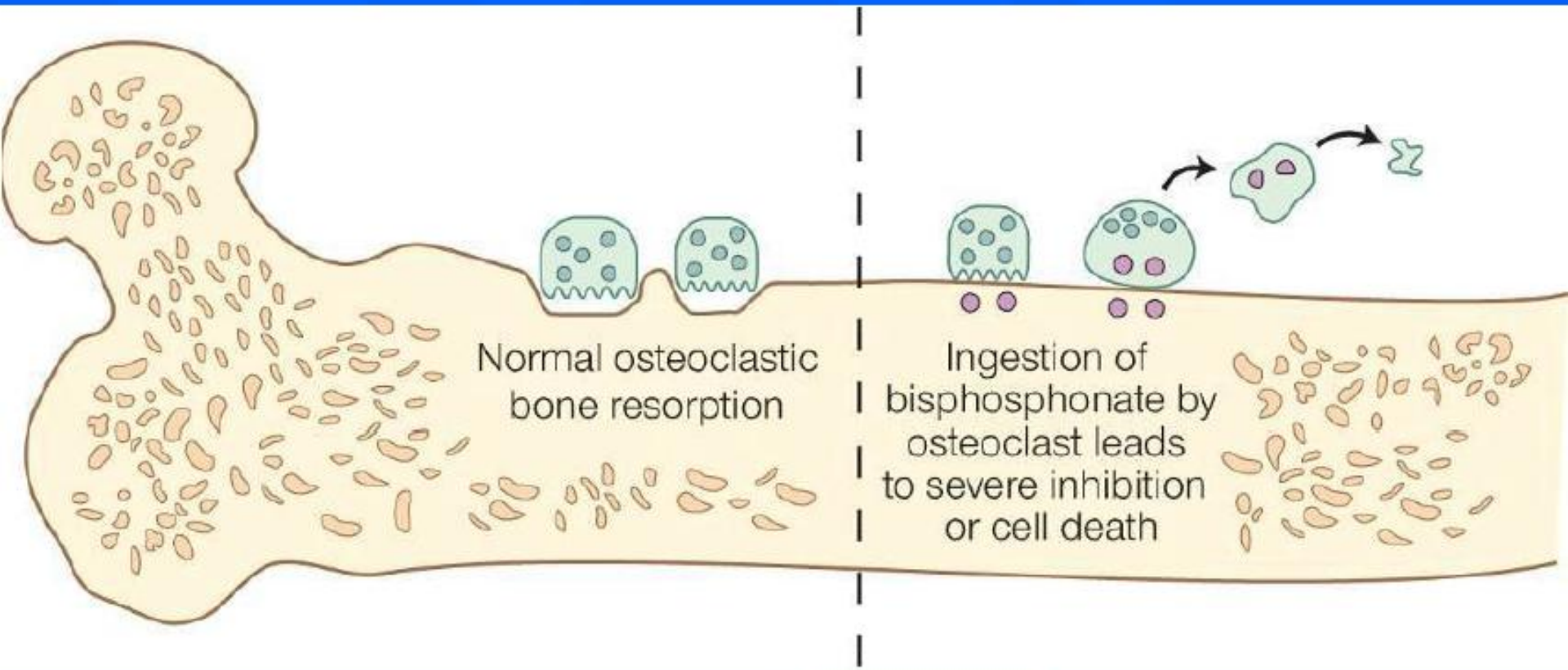
- Solid tumours metastatic to bone, multiple myeloma
- ↓SREs –hypercalcemia, new bone mets, bone pain, pathologic #2° to bone metastases

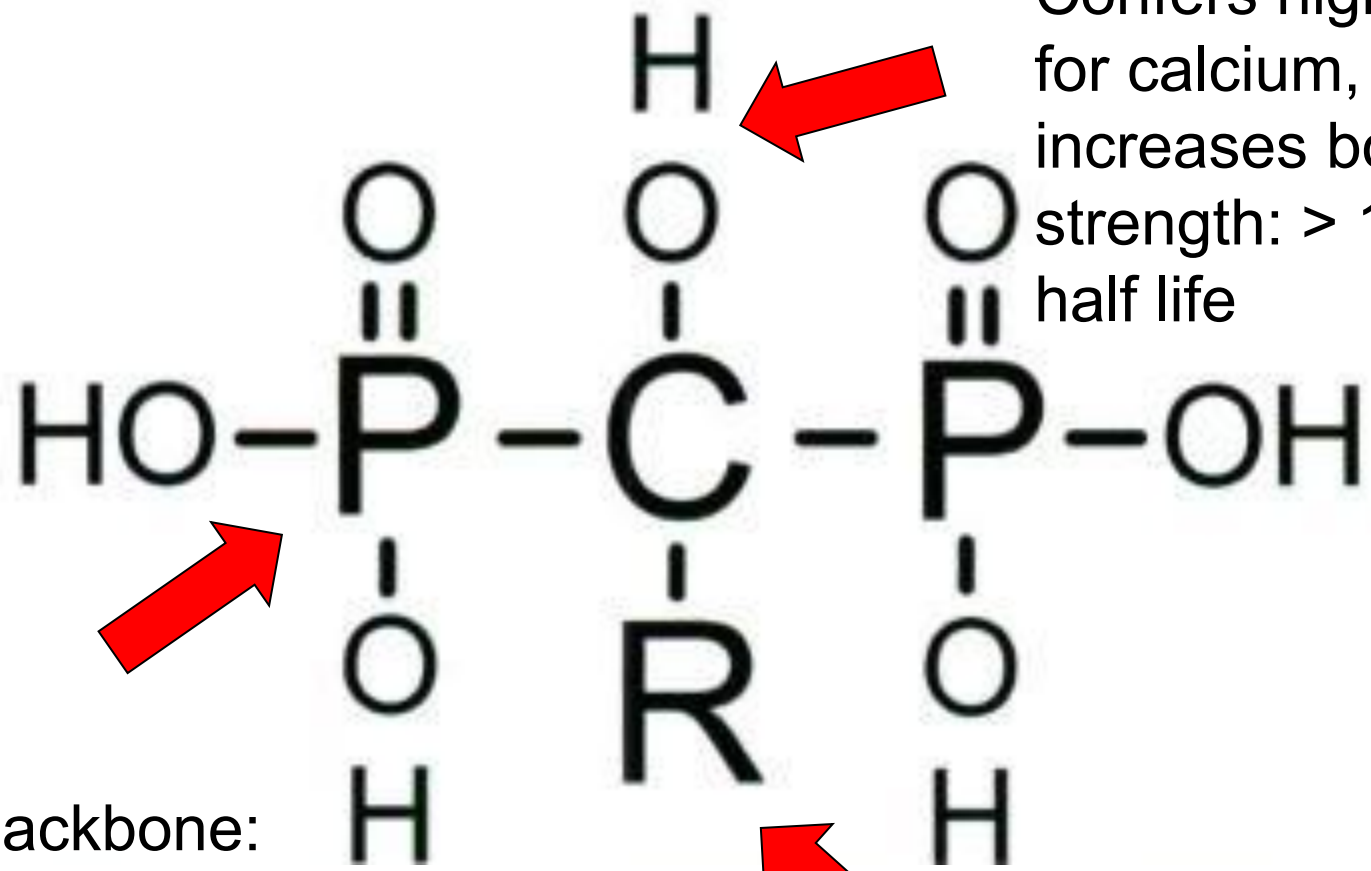
Mechanism of Action

- Osteoclast death at resorption sites.
- Inhibition of osteoclast precursors in bone marrow.

Mechanism of Action

- strong affinity for bone at sites of active remodelling
- potent inhibitors of resorption and remodeling activity
- pyrophosphate-like chemical substructure that confers a strong affinity for calcium and bone tissue
- inhibits enzymes that influence osteoclast activity
- ? anti-angiogenesis at high doses


Normal osteoclastic
bone resorption
Ingestion of
bisphosphonate by
osteoclast leads
to severe inhibition
or cell death



Confers high affinity for calcium, increases bond strength: > 10 year half life

PCP backbone: BP unable to be hydrolyzed; irreversible binding to HA crystals

Determines potency. Nitrogen-containing side chains most potent

Classes

Non-nitrogen containing

- Lower potency, generate cytotoxic analogs of ATP → interfere with mitochondrial fxn → apoptosis of osteoclasts
- Clodronate, tiludronate, etidronate

Classes

Nitrogen containing

- Highly potent, inhibit mevalonate cellular metabolic pathway
- Pamidronate, zoledronate, alendronate, risedronate, ibandronate

Route of Administration

Intravenous

- entire dose not cleared by kidney (60-70%) rapidly bound to bone
- risk ONJ at 5 doses of therapy

Oral

- ~0.64% absorbed, additional 30-40% cleared by kidneys
- many more repetitive doses to accumulate equal level in bone
- risk ONJ at 3 years or longer

Dose (Oral)

- Alendronate (OP) 70 mg/wk.
- Residronate (OP) 35mg/wk.
- Ibandronate (OP) 150 mg/month.
- Clodronate (OP) 1600 mg/day.
- Etidronate (Paget's) 500 mg/day.
- Tilidronate (Paget's) 400 mg/day.

Dose (IV)

- Zolendronate (OP) 5 mg/yr.
- Zolendronate (Cancer/Mets) 4 mg/month.
- Pamidronate (Cancer/Mets) 90mg/month.

RANKL Inhibitor Mechanism of Action

- Osteoclast inhibition at resorption sites.
- Osteoclast inhibition in blood and tissue spaces.
- Osteoclast precursor inhibition in bone marrow.

RANKL inhibitors (Subcut)

- Denosumab (OP) 60mg/month.
- Denosumab (Cancer/Mets) 120 mg/month.
- Xgeva and Prolia.

Bisphosphonates vs. RANKL Inhibitors

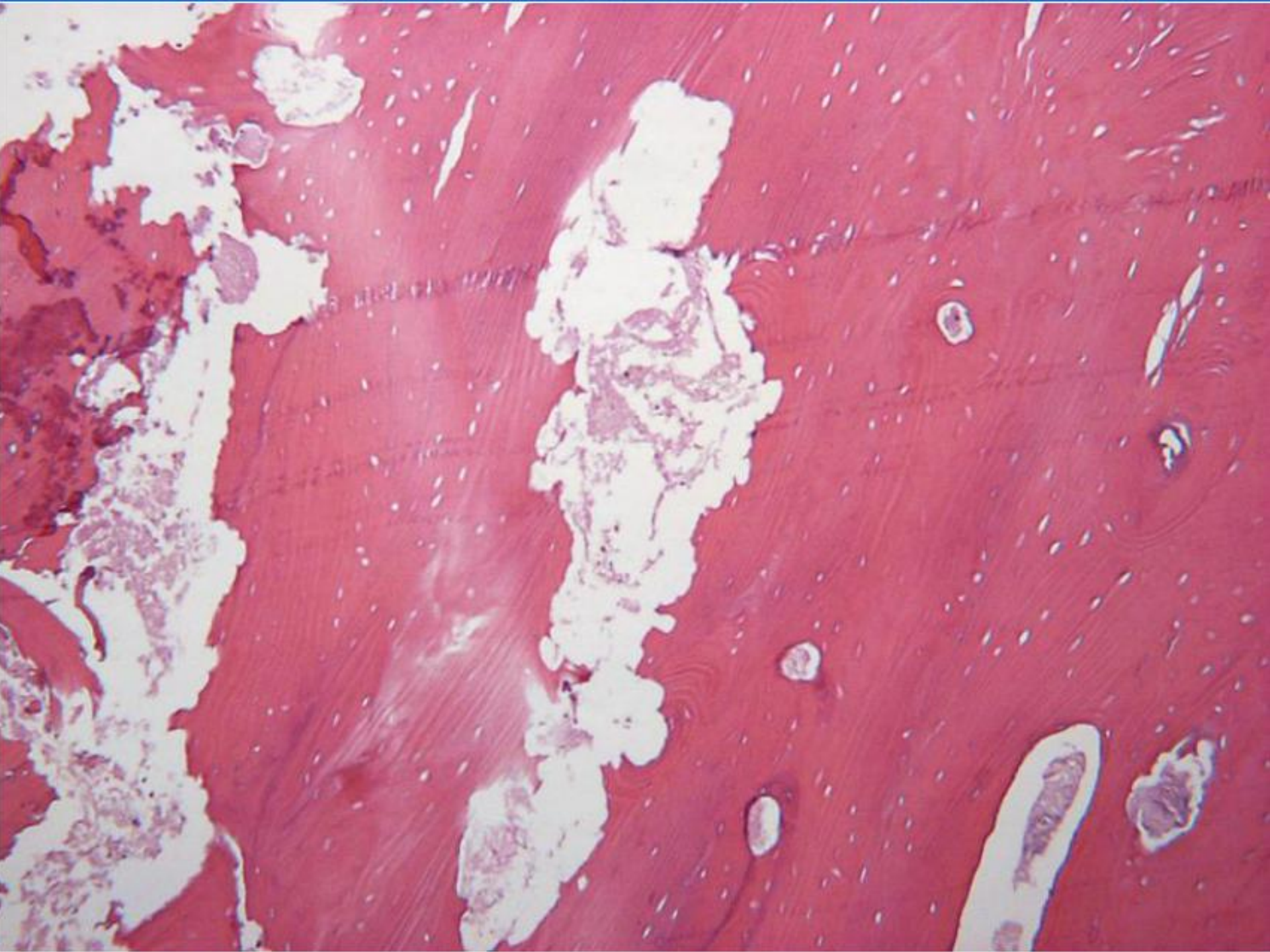
- Most Bisphosphonates affect osteoclasts resorbing bone and only some affect precursors in marrow.
- RANKL Inhibitors affect mature osteoclasts and their precursors everywhere.

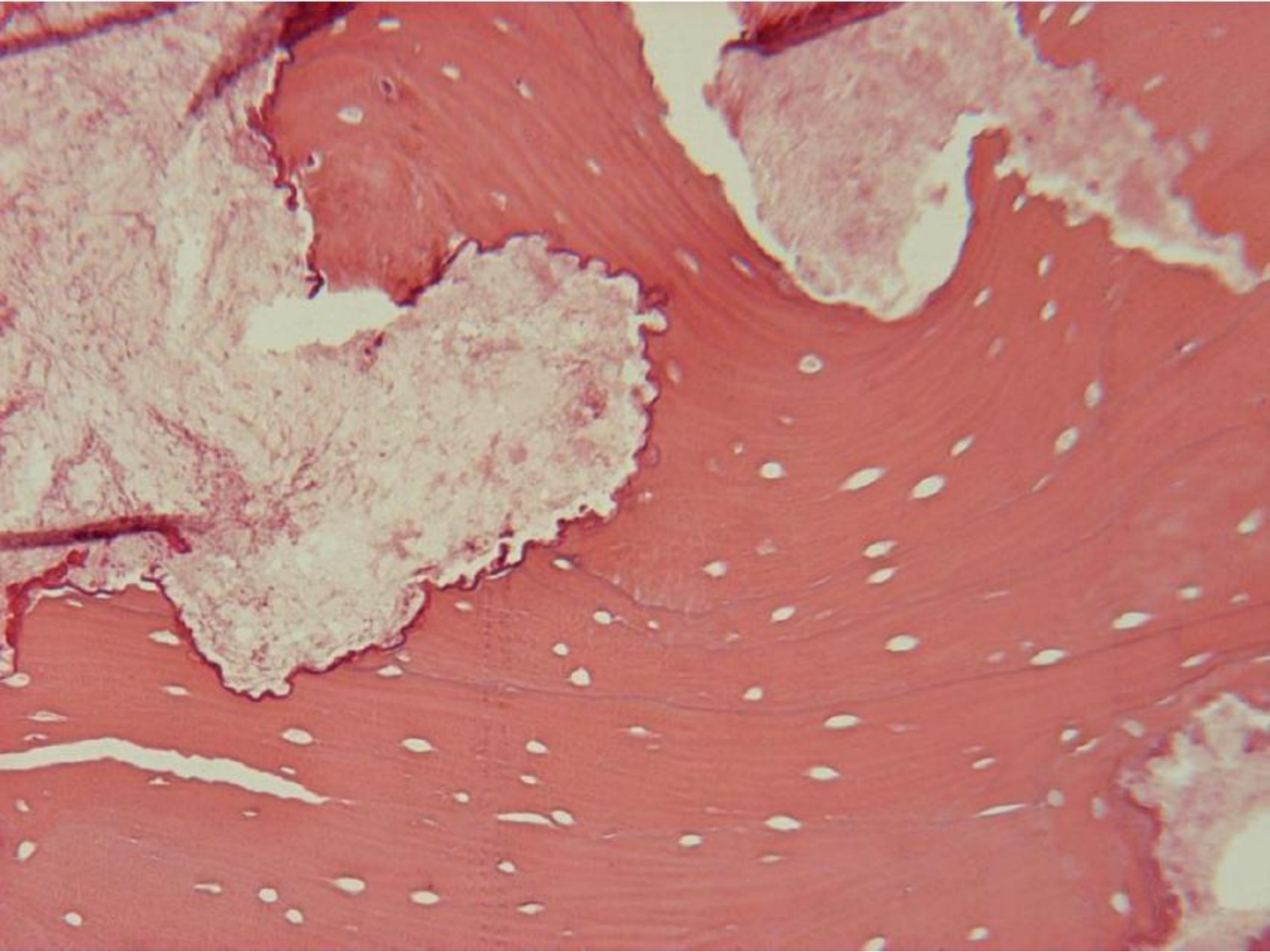
Anti-angiogenic Drugs (IV)

- Bevacizumab (Cancer/Mets) 500mg/2 weeks.
- Sunitinib (Cancer/Mets) 25 mg/day.

Pathogenesis

- profound inhibition of osteoclast function inhibits normal bone turnover
- defect in physiologic remodeling and wound healing
- selective involvement of the maxilla and mandible may be a reflection of the unique environment of the oral cavity:
  - bones with high turnover rates
  - continuous micro trauma from chewing
  - oral bacterial load





BRONJ Staging (AAOMS)

- Stage I: asymptomatic exposed bone.
- Stage II: exposed bone, pain +/- infection.
- Stage III: as above plus one or more of the following:
  - Pathologic fracture.
  - Extra-oral fistula.
  - Osteolysis extending to inferior border mandible.

3D D= n/a
ed D=18.26 mm
3D D=18.26 mm
ed D=0.00 mm
3D D= n/a
1
2

MRONJ – Differential Diagnosis

- Herpes Zoster.
- Osteoradionecrosis.
- Osteopetrosis.
- Osteomyelitis.
- Cemento-osseous dysplasia.

MRONJ - Microbiology

- Actinomyces.
- Veillonella.
- Eikenella.
- Moraxella.

Incidence

American Society for Bone and Mineral Research
JBMR 22:10, 2007

- Malignancy 1-10%
- Osteoporosis 1/10,000-1/100,000

- Information on incidence rapidly evolving
  - data to date is retrospective: reports missing important historical and clinical information; poor definition of the disease
  - underreporting of true cases
  - recent adoption of BP therapy in some countries
  - postmenopausal women switching from estrogen replacement to BP therapy

MRONJ – Important Considerations

- Risk factors.
- Initiating factors.
- Vulnerable sites.
- Co-morbidities.

Risk Factors

- Potency.
- Dose.
- Frequency of administration.
- Route of administration.
- Half-life.
- Duration of drug use.

Risk Factors

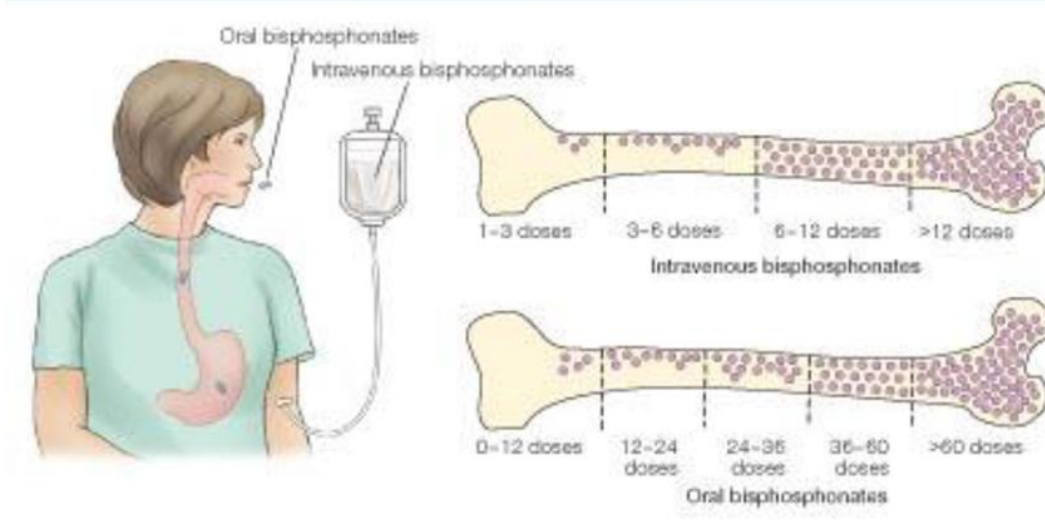
- MRONJ is due to the accumulation of toxic levels in bone.

Risk factors

- Oral MRONJ cases:
  - Fosamax >> Actonel.
- Fosamax 70 mg/wk.
- Actonel 150 mg/month.

Route of Administration

- IV dose has 140 times the bone bioavailability of an oral dose.



Half life

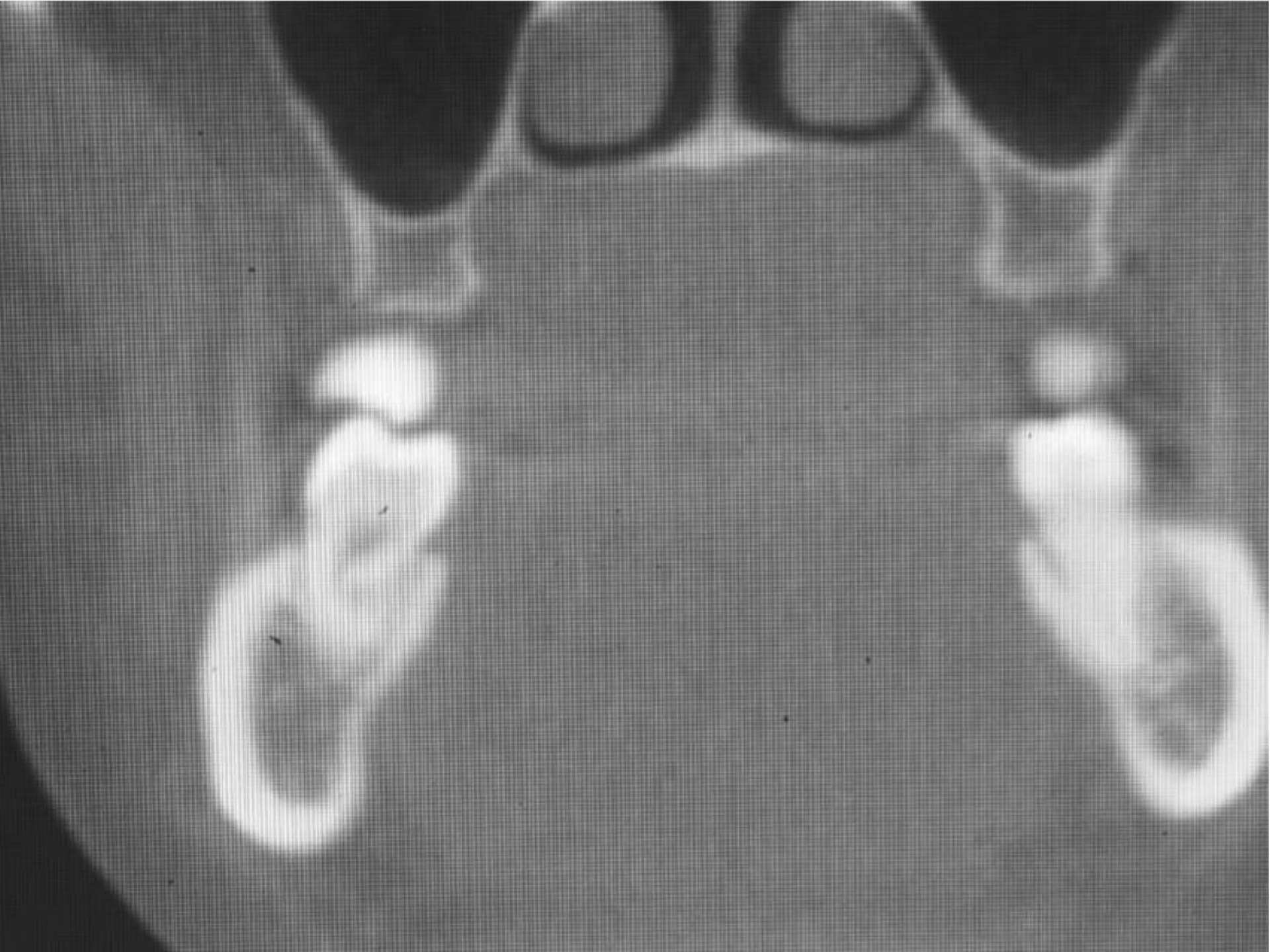
- Bisphosphonates: 10+ years.
- Denosumab: 26 days.

Initiating Factors

- Spontaneous.
- Dental extractions.
- Alveolar bone surgery.
- Traumatic occlusion.

Vulnerable Sites

- Tori.
- Alveolar bone.
- mandible>maxilla.
- Posterior mandible (esp. lingual).



Neviaser MD et al: Low-energy femoral shaft fracture associated with Alendronate use

J Orthop Trauma 22:346-351, 2008.

A

Co-morbidities

- Drugs (steroids, chemotherapy).
- Diabetes.
- Smoking.
- Periodontal disease/poor oral hygiene.
- Etc.

Prevention and Management

"So, it's unanimous then – nobody has a clue what to do."

Canadian Guidelines

Canadian Consensus Practice Guidelines for Bisphosphonate-Associated Osteonecrosis of the Jaw[1]

June 2007

[1]Khan et al. Journal of Rheumatology. 35(7):1391-7, 2008 Jul.

American Task Force

Bisphosphonate-associated osteonecrosis of the jaw: Report of a task force of the American Society for Bone and Mineral Research[1]

[1]Khosla et al. J Bone and Mineral Research 2007; 22(10):1479-91

AAOMS 2009

Diagnosis

- Presence of exposed bone in the maxillofacial region for longer than 8 weeks
- Past or present BP therapy
- No history of radiotherapy to the jaws
- Radiographic findings non-specific
- Serum telopeptide markers of bone turnover (CTx) under study

Prevention

- Appropriate screening and elimination of dental disease prior to start of therapy.
- Avoidance of elective oral surgery once therapy started.
- Discontinuation of drug if surgery essential.

Management

General Principles

- Appropriate pain control
- Treatment of frank infection
- Conservative surgical management
- Nutritional support as needed

Management

Stage 1: asymptomatic, no evidence of infection

- Baking soda mouth rinse
- Follow up every 3 months
- Patient education and review of indications for continued BP therapy

Stage 2: pain, clinical evidence of infection, such as local erythema, ± purulent drainage

- Symptomatic treatment with broad-spectrum oral antibiotics
- Antibacterial mouth rinse
- Pain control
- Only superficial debridement to relieve soft tissue irritation

Stage 3: pain, infection, one or more of the following: pathologic fracture, extra oral fistula, osteolysis extending to inferior border of the mandible

- Antibacterial mouth rinse
- Antibiotic therapy
- Pain control
- Surgical debridement/resection for longer-term palliation of infection and pain
- Dietary supplementation or tube feeding if necessary
- HBO not effective

AAOMS 2014 Update

Prevention of Osteonecrosis

- Recommendations before therapy starts.
- Recommendations during/after therapy.

Recommendations before therapy

- Prophylaxis.
- Treat caries.
- Treat periodontal disease.
- Extract hopeless teeth.
- Excise tori? Multilobed?
- Occlusal adjustments.
- Defer bisphosphonate therapy for two months.

Recommendations during therapy

- Supragingival scaling.
- Treat caries; if needed root canal and amputate crowns.
- Splint mobile teeth.
- Avoid invasive procedures including implants.
- If extractions are unavoidable, make sure patient is informed of risks.

Treatment of Osteonecrosis

- Avoid debridements.
- Smooth sharp edges of bone.
- Antibiotics.
- Surgery.

Treatment of Osteonecrosis

- Antibiotics – pain from secondary infection (necrotic bone is not itself painful).
- Drugs:
  - Penicillin VK 500/600 mg qid.
  - Peridex.
  - If Pen allergic: Levaquin or doxycycline.
  - Add Flagyl 500 mg bid x 10 days for refractory cases.

Treatment of Osteonecrosis

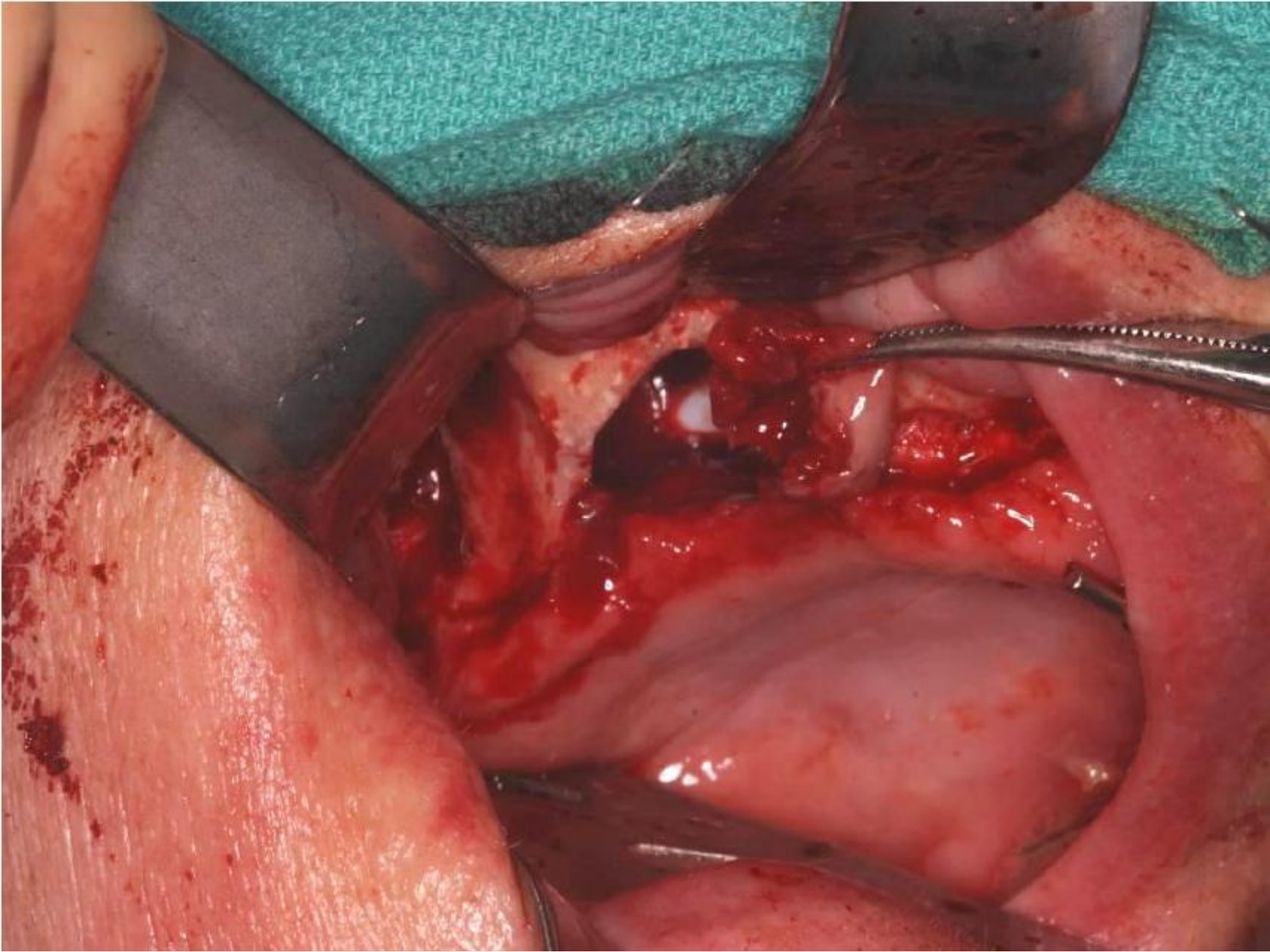
- Ineffective therapies:
  - Clindamycin.
  - Hyperbaric oxygen.
  - Ozone.
  - Laser.

Treatment of Osteonecrosis - Surgery

- Indications:
  - Symptomatic cases refractive to non-surgical therapy.
  - Pathologic fracture.
  - Direct sinus communication.

Treatment of Maxillary Osteonecrosis - Surgery

- Sequestrectomy of little value.
- Alveolar resection usually inadequate.
- Posterior maxillary resection with sinus debridement usually sufficient.
- Buccal fat pad and mucosal closure.
- Buccal fat pad and floor of nose have rich blood supplies.



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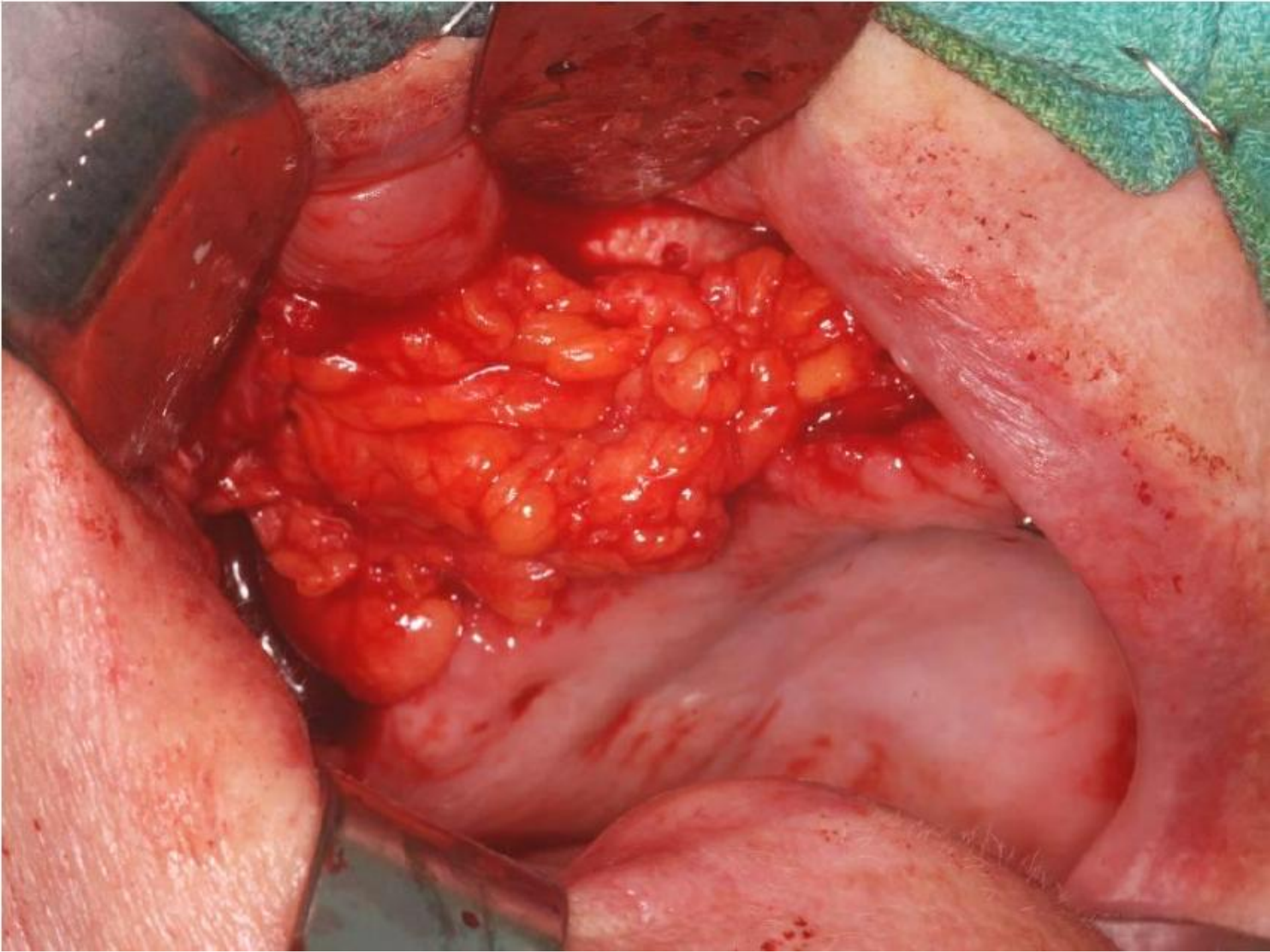
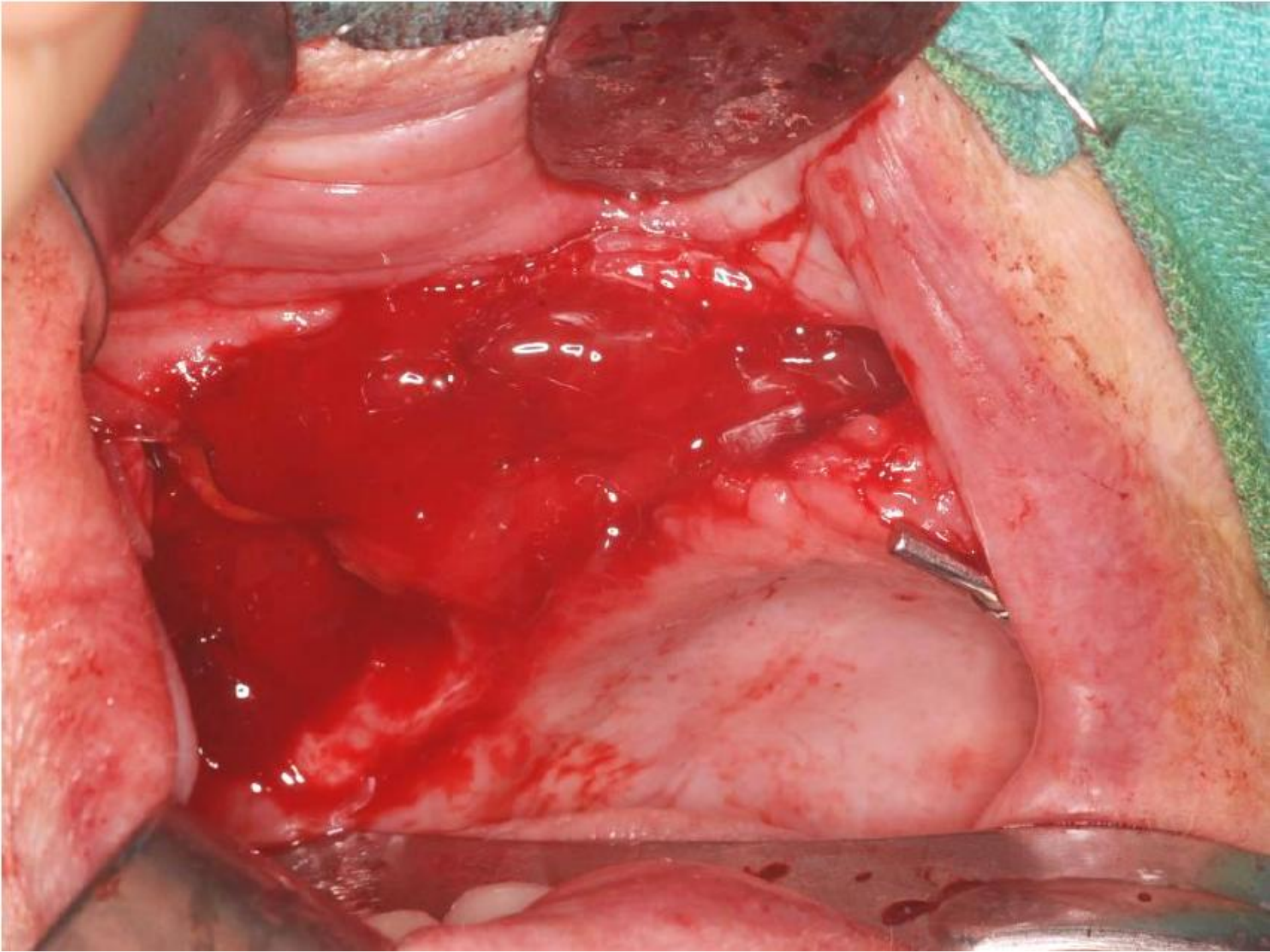
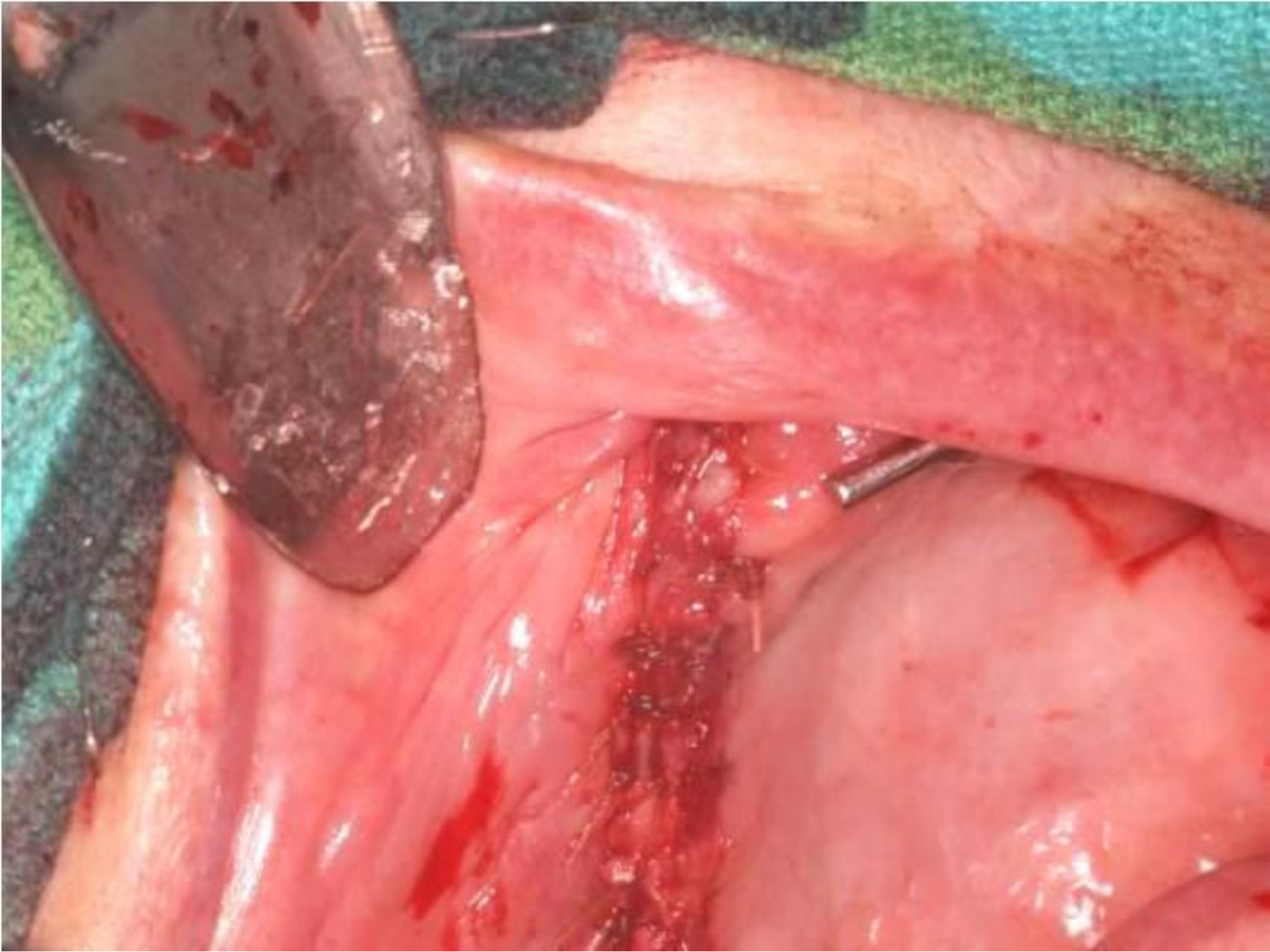
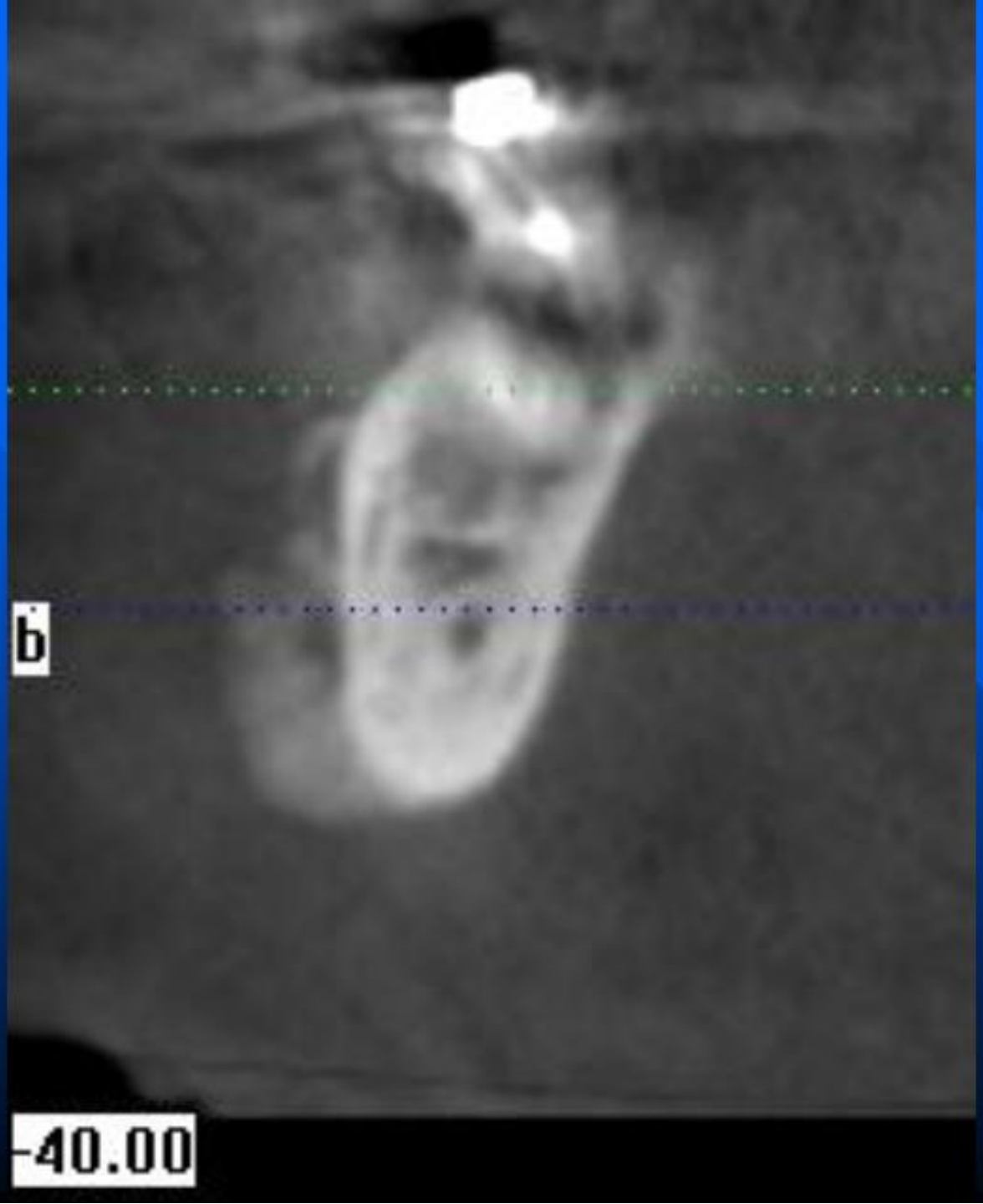






Treatment of Mandibular Osteonecrosis - Surgery

- Sequestrectomy little or no value.
- Alveolar resection usually inadequate.
- Continuity resection required:
  - A palliative procedure.
  - Resect to low risk bone (ramus and incisor region: both have decreased occlusal loads).
  - Look for bleeding bone; residual marrow.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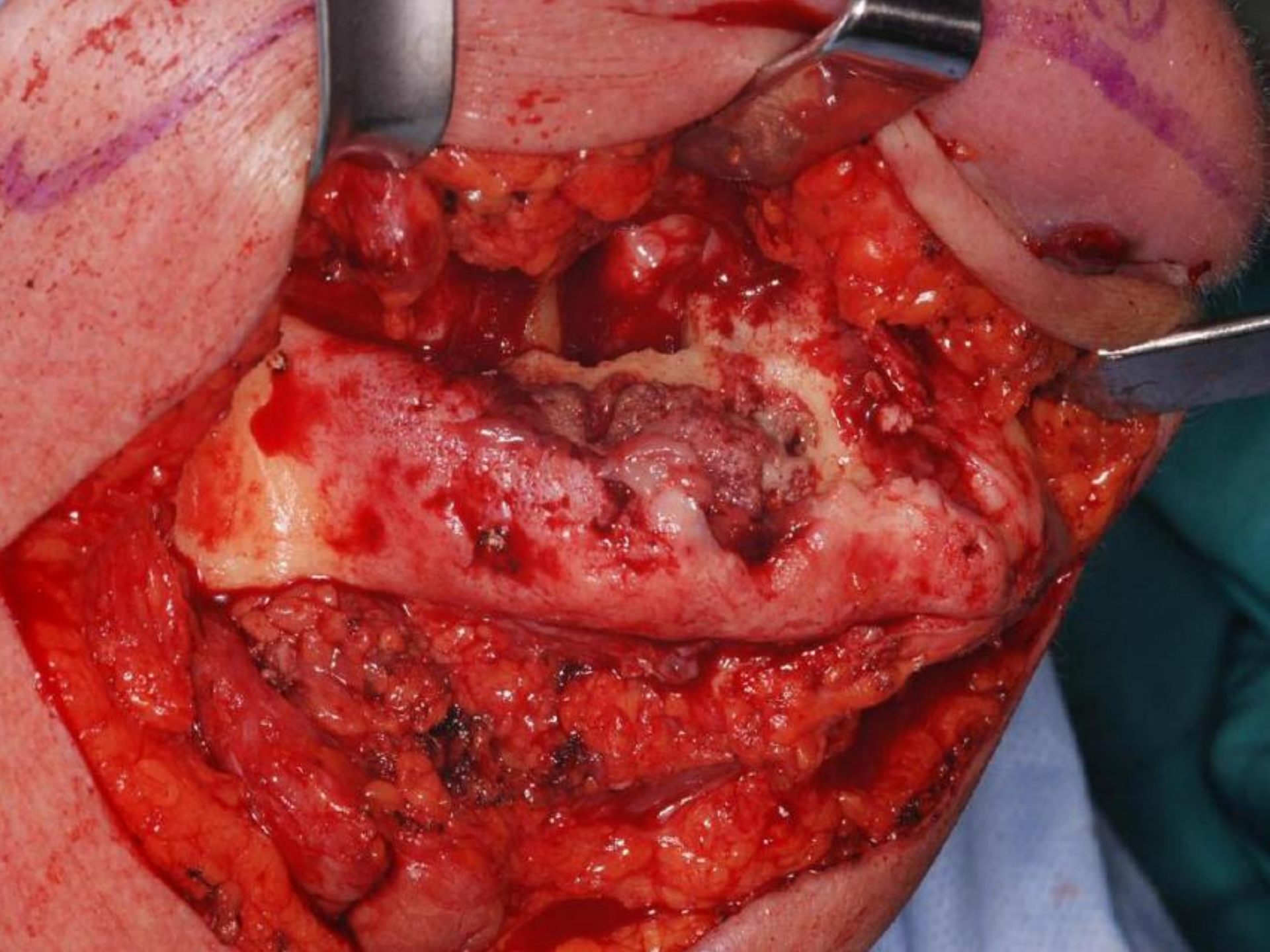
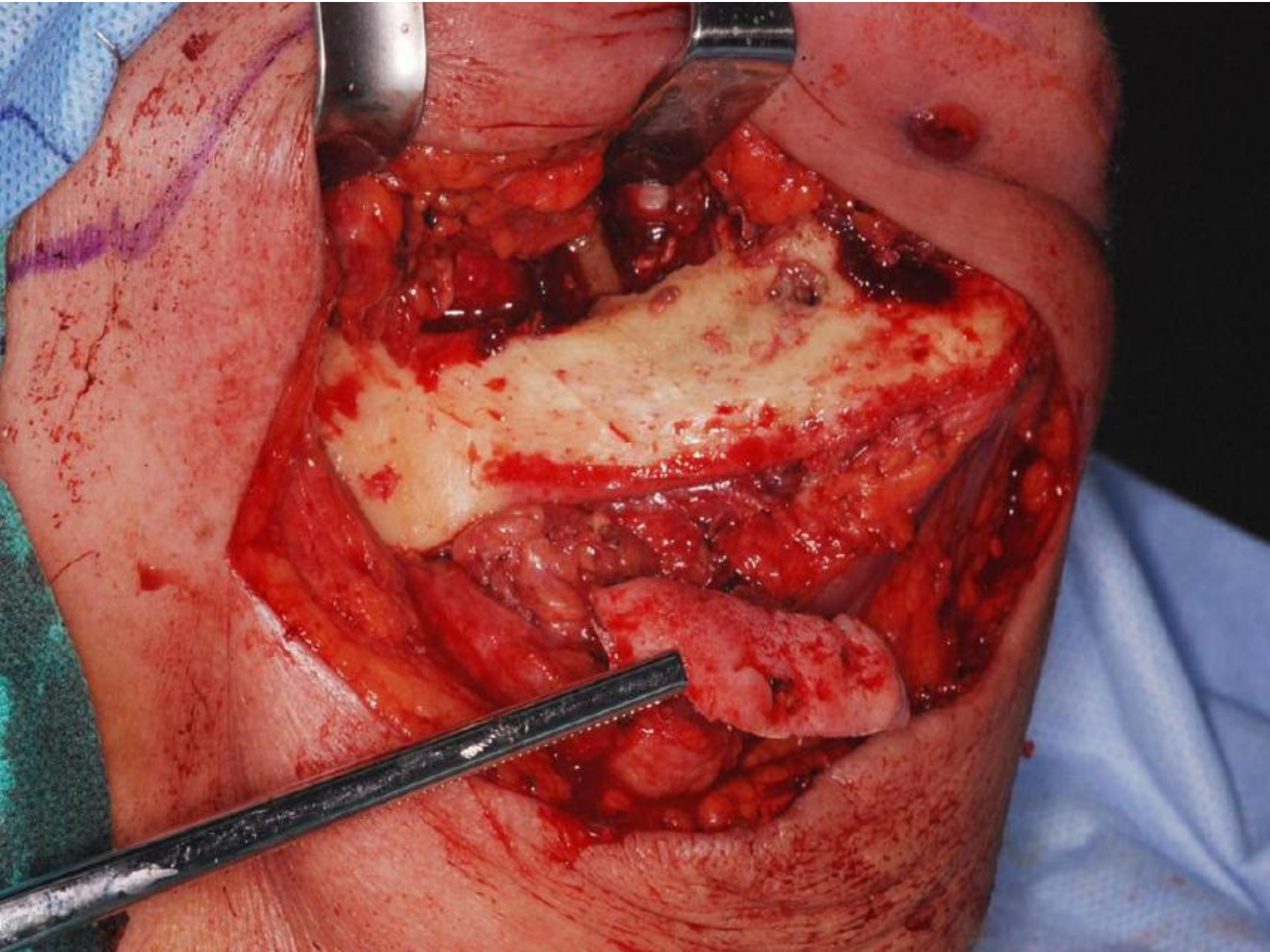
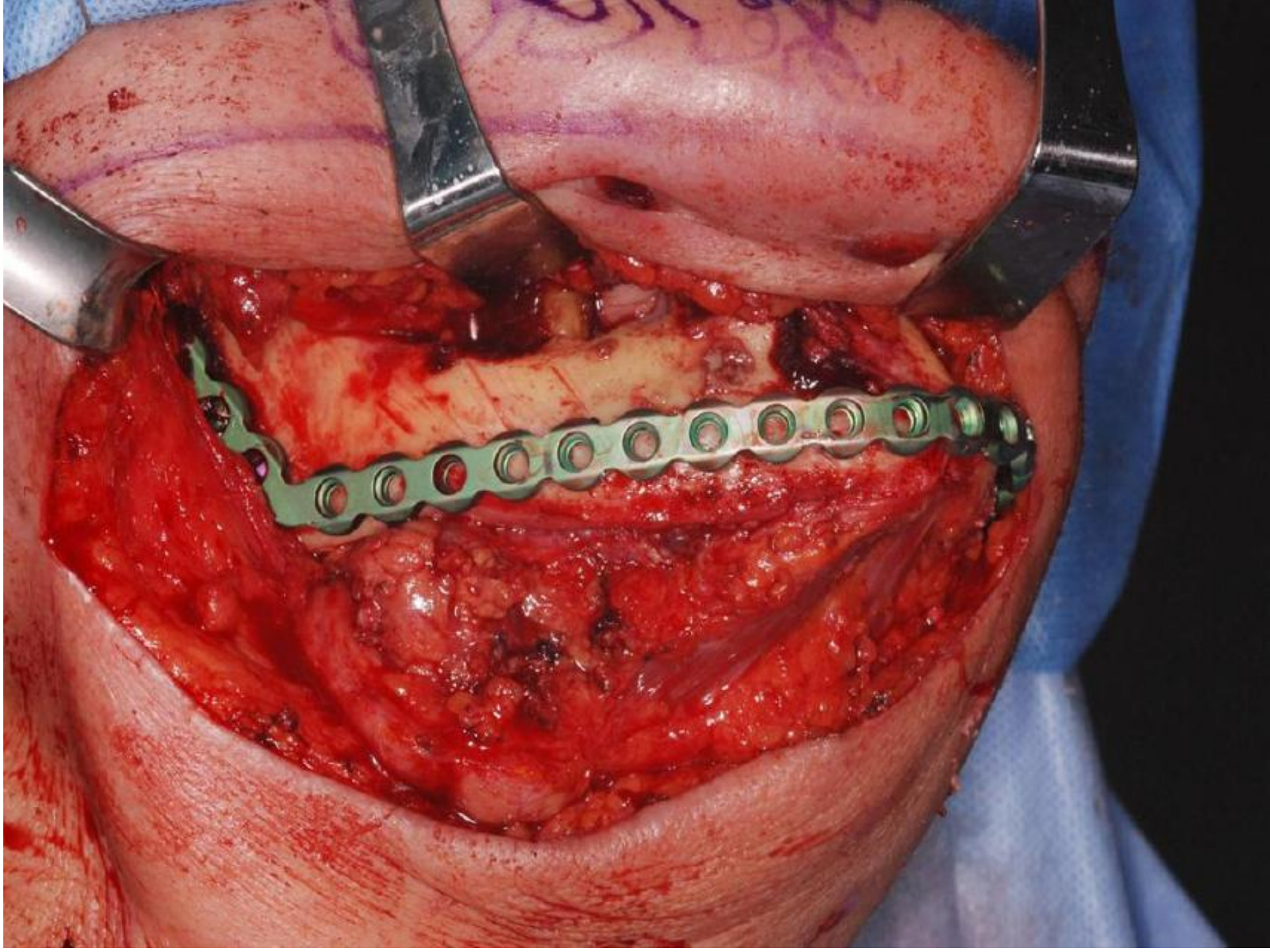






Treatment of Osteonecrosis

- Drug holiday:
  - Will not resolve MRONJ.
  - May assist with wound healing.
  - May reduce the likelihood of a second MRONJ focus.

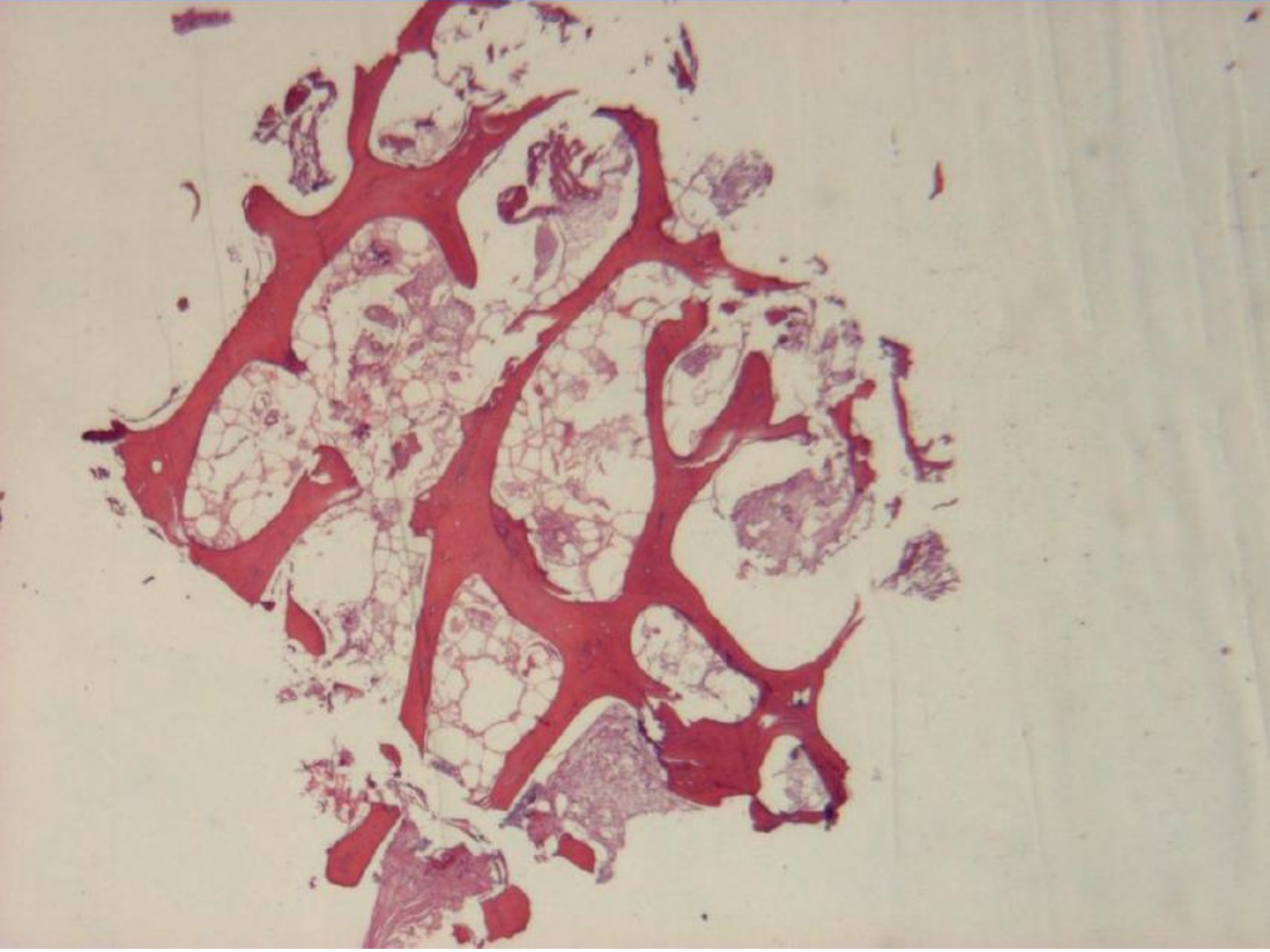
Treatment of Osteonecrosis

- Bony reconstruction is not possible:
  - There is no disease free donor bone.
  - BMP’s are contraindicated in cancer patients.

The Osteoporosis Patient

Osteoporosis

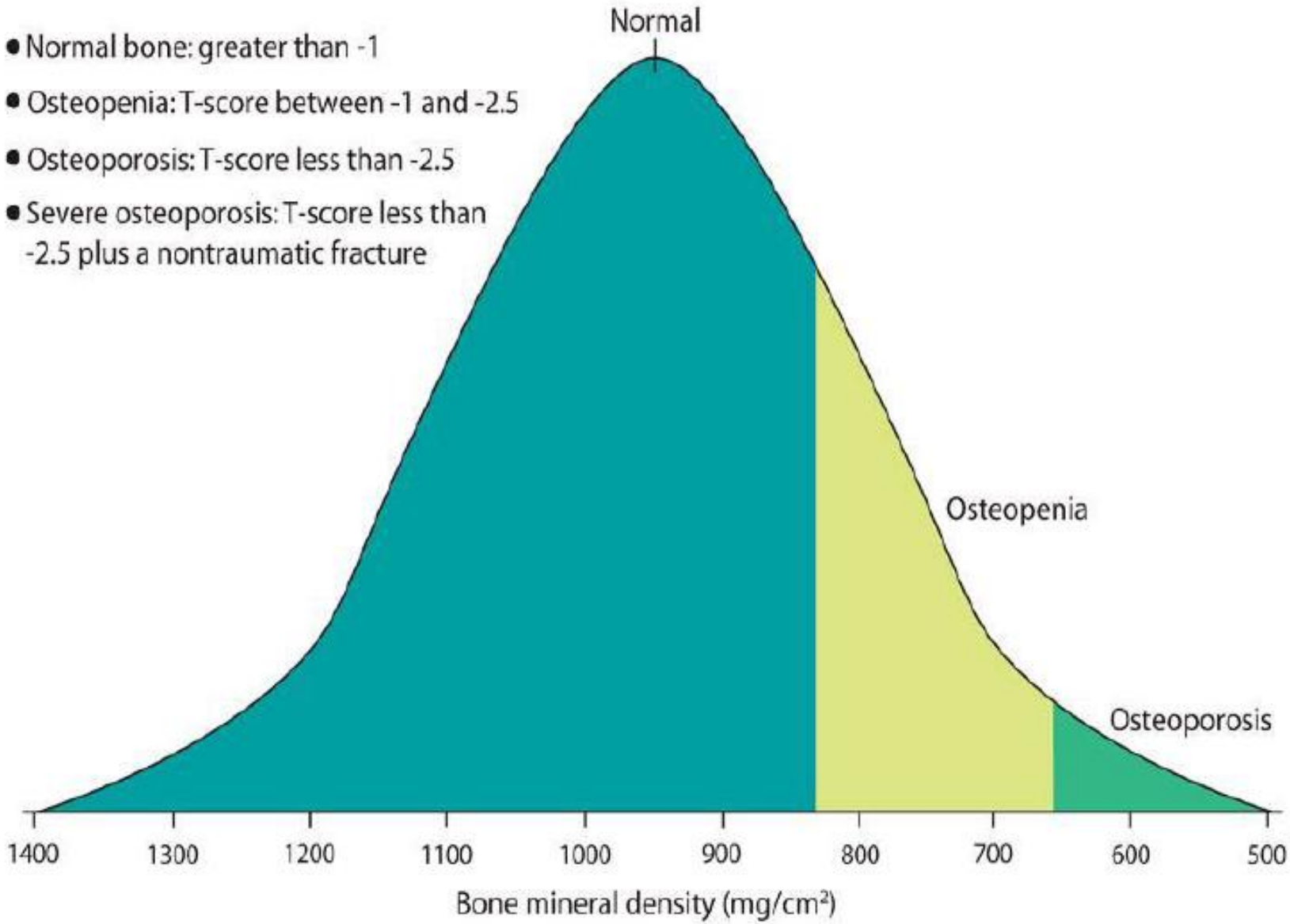
- Weakening of bone due to:
  - Thinning of trabeculae.
  - Thinning of cortex.



Osteoporosis

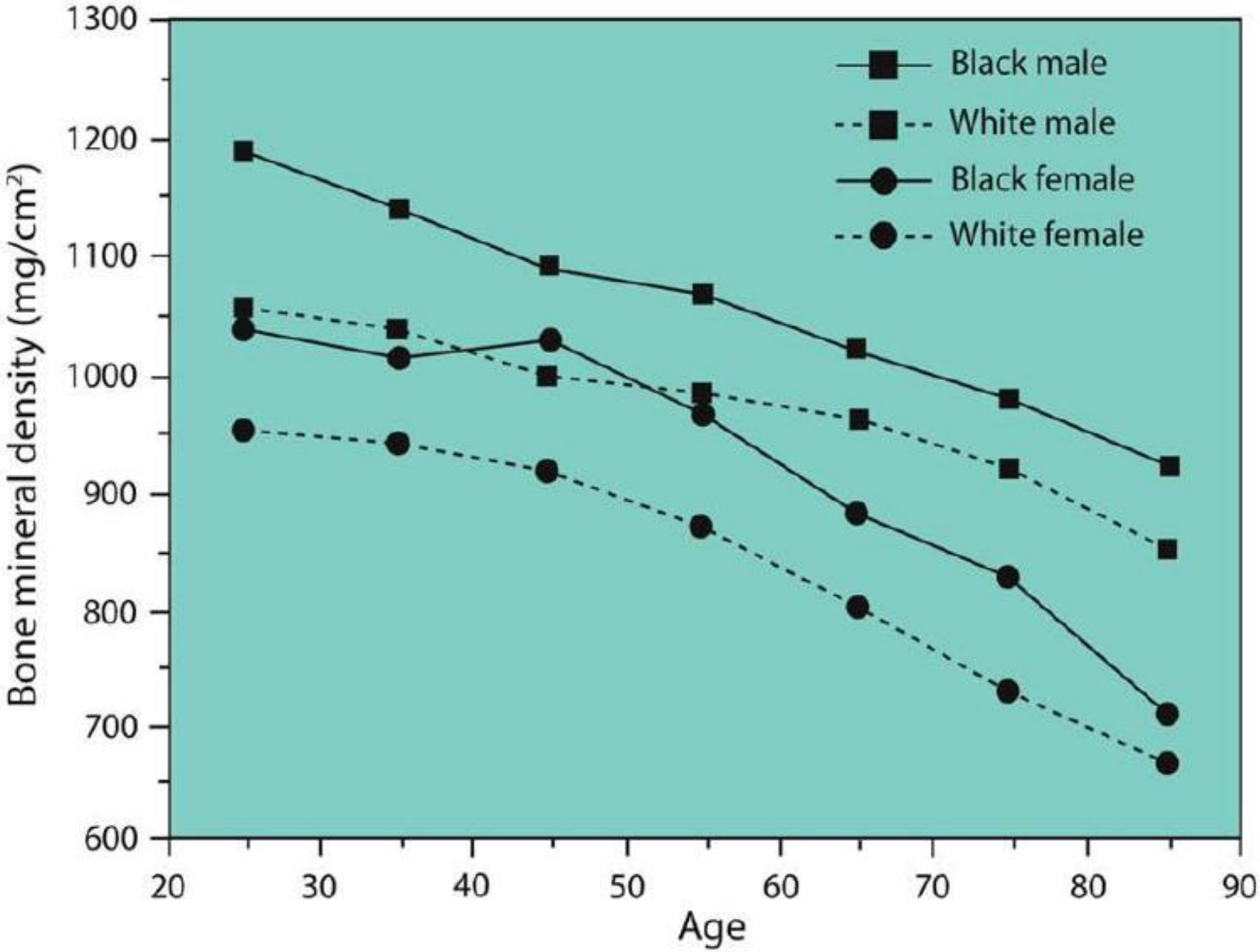
- According to the WHO:
  - Osteopenia is not a disease.
  - Osteoporosis is a disease.
  - Bone mineral density is used as the marker.
  - Osteoporosis is defined as T score < minus 2.5.

1.04 g/cm2
0.22 g/cm2


• Normal bone: greater than -1
• Osteopenia: T-score between -1 and -2.5
• Osteoporosis: T-score less than -2.5
• Severe osteoporosis: T-score less than -2.5 plus a nontraumatic fracture
Normal
Osteopenia
Osteoporosis
1400
1300
1200
1100
1000
900
800
700
600
500
Bone mineral density (mg/cm²)

Osteoporosis

- Bone mineral density is not necessarily a good marker for disease:
  - Inter-device variation.
  - Inter-operator variation.
  - A denser bone is not necessarily a stronger bone.


Black male
White male
Black female
White female
Bone mineral density (mg/cm²)
1300
1200
1100
1000
900
800
700
600
20
30
40
50
60
70
80
90
Age

Osteoporosis

- Treatment focuses on preventing skeletal events (ie. fractures).
- Treatment options:
  - Anti-resorptive drug therapy.
  - Nutrition/exercise/fall prevention regimens.

Osteoporosis – Drug therapy

- Pharmaceutical companies claim:
  - Osteoporosis begins in childhood and affects everyone.
  - 50% of women will fracture a bone.
  - Considerable mortality accompanies hip fractures.
  - Enormous cost to health care system.

Osteoporosis – Drug Therapy

- Bisphosphonates inhibit bone turnover/resorption, resulting in increase bone density.

Osteoporosis – Non drug therapy

- Drug therapy stats are inflated.
- Bones fracture because of poor bone quality.
- Fractures occur because of falls, not osteoporosis.
- Bisphosphonate therapy results in older, more brittle bones.

Current recommendations

- The FDA has reviewed several recent studies that cast doubt on the benefits of bisphosphonate therapy and in 2011 released the following guidelines:
  - Every woman on bisphosphonates for osteoporosis must be examined after 3 years.
  - No one on bisphosphonates for osteoporosis should be taking them for more than 5 years.

Osteoporosis – New therapies

- Calcium and Vit D3
- Raloxifene (Evista).
- rhPTH1-34 (Forteo).

Prevention of Osteonecrosis

- Non-invasive dentistry is safe at all times.
- Oral bisphosphonate 9 month drug holiday (and CTX > 150 pg/ml).
- Prolia 3 month drug holiday.
- Reclast 9 months since last dose; delay next dose 3 months post-op.

Prevention of Osteonecrosis

- The recovery from bisphosphonate bone turnover suppression is delayed by:
  - Steroids.
  - Methotrexate.
  - Greater than 8 year use.
- Consider 12 month drug holiday.
- All drug holidays should extend to 3 months post-op.

Orthodontics

- Teeth will not move in a patient on bisphosphonates or Denosumab.

Treatment of Osteonecrosis

- Drug holiday.
- Debridement.
- Reconstruction including dental implants.

CTx

- Specific marker of bone turnover and therefore healing.
- Usefulness in predicting osteonecrosis risk is controversial.

CTx

- Only useful for patients on bisphosphonates:
  - < 100 pg/ml high risk.
  - 101-150 pg/ml mod risk.
  - > 151 pg/ml low or no risk.

CTx

- Cannot be used for patients:
  - With cancer (value will be artificially high).
  - On steroids (value will be artificially low).
  - On methotrexate (value will be artificially low).